

National University of Computer and Emerging Sciences



Lab Exercise 09 Data Analysis & Visualization

Course Instructor	Umme-Amara
Lab Instructor (s)	Gullsher Chaudhary Rida Mahmood
Section	
Semester	Fall 2023

Department of DataScience
FAST-NU, Lahore, Pakistan

References:

Lecture 17 of Theory RNN Concepts

RNN (Recurrent Neural Network)

Implement RNN from scratch using following steps;

1. Importing the libraries & load data set
2. Data Preprocessing & Training
 - a) Tokenize Text
 - i. convert in sentences
 - ii. convert in words
 - b) Remove infrequent words
 - i. remove stopwords
 - c) Build training & test dataset
3. Building the RNN
 - a) Initialize Assistance parameters
 - i. word_dim, hidden_dim, output_dim, bptt_truncate
 - b) Initiaize Network parameters
 - i. U, V, W
 - c) Activate Function
 - i. Sigmod
 - d) Forward_Propagation
 - i. do forward pass to get prediction
 - ii. Prodict the highest score
 - e) Calculate loss
 - i. create loss function to measure the errors

Note:

Evaluating loss depends on the size of the dataset, so try to minimize it by random.